Hi-Fi Prototype

Team 7

Team Members

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‘26

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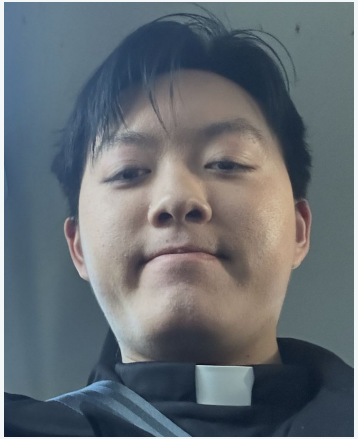

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Reason Lab
Reflect. Learn. Grow.

Project Name: Reason Lab

“Better Learning Starts With You “

Value Proposition

Reason Lab enforces active learning by requiring students to attempt reasoning before receiving assistance. Instead of giving instant answers, the system provides adaptive hints and structured feedback to build durable conceptual understanding

Problem Overview

- Audience / Users
  - College + High School Students dealing w/ complex and difficult to understand coursework

- Problem
  - Nowadays with AI so readily available, many are using it as a shortcut.
  - This use of AI is taking away from in-depth learning, and can hinder critical thinking

What is our solution?

- The value proposition of Reason Lab is to enforce active learning by requiring users to attempt a solution before receiving assistance.
- Solution:
  - Makes its users strengthen their own reasoning rather than an instant solution
  - Will give adaptive guidance and guide the user, forcing a more durable understanding of material
  - Through adaptive, visual guidance, ReasonLab responds to user attempts and misconceptions, helping learners refine their thinking and build a more durable conceptual understanding over time.

Values in Design

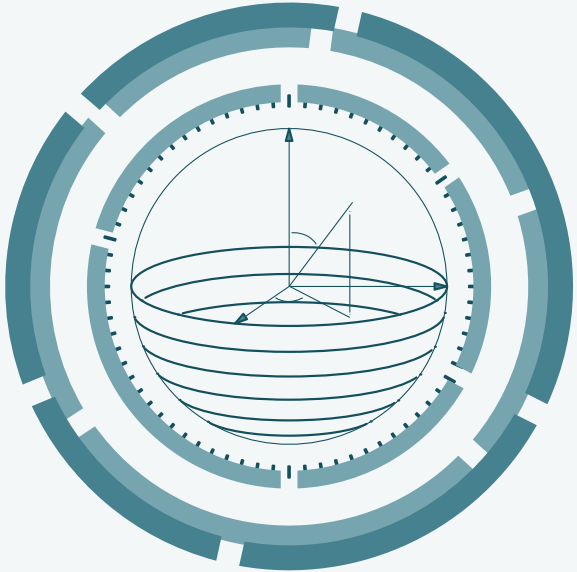
Active Learning

AND

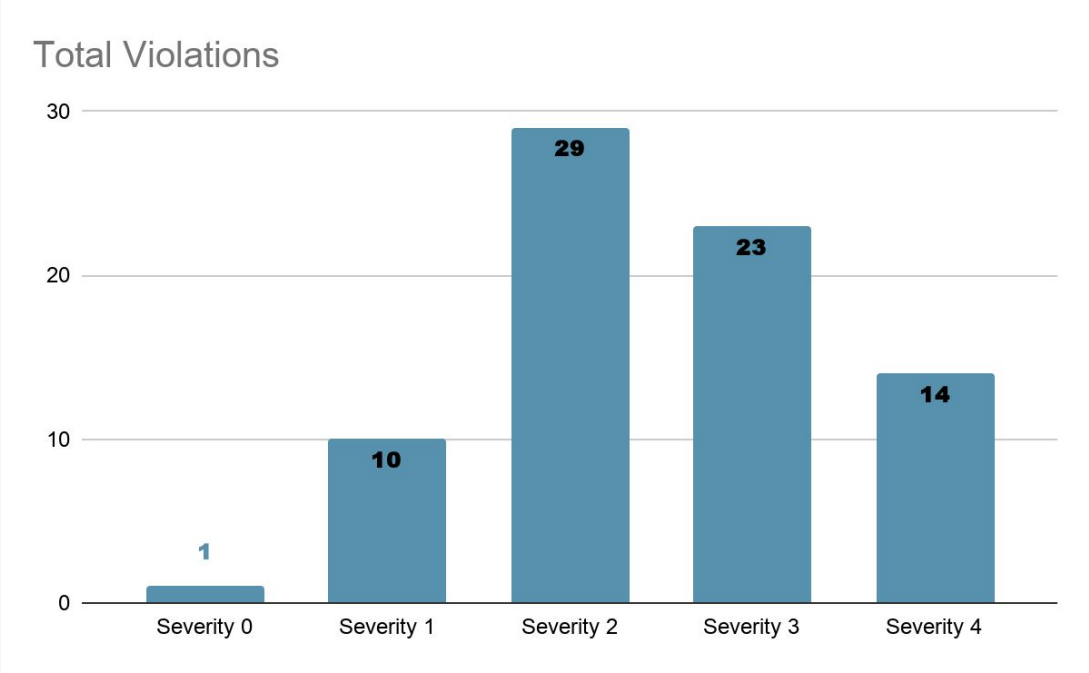
User Control

01

Heuristic Evaluation Results



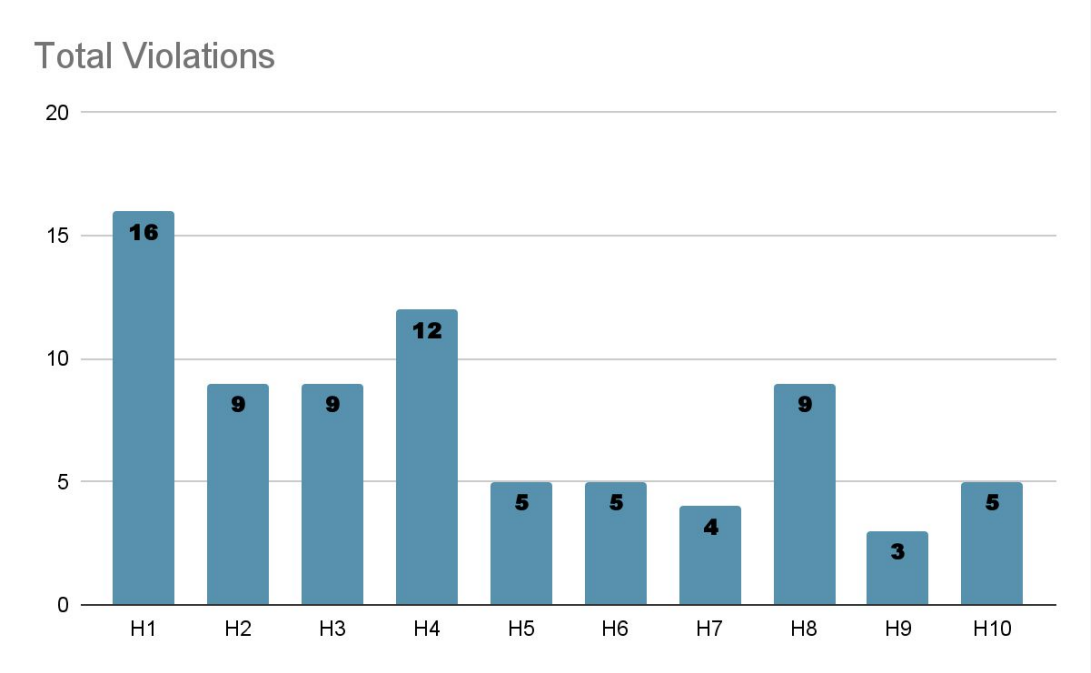
Violations by Severity



Violations by Heuristic

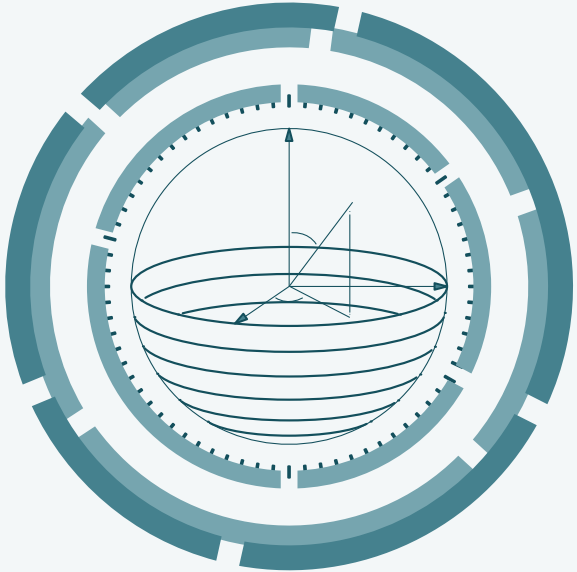
Biggest Pain Points:

- H1: Visibility of System Status
- H4: Consistency & Standards



02

Key Violations & Insights



Heuristic Evaluation Key Insights

77

Total Violations Found

Severity Breakdown

S4 Critical	14
S3 Major	23
S2 Moderate	29
S1 Minor	10

Most Cited Heuristics

H1 — Visibility of System Status
Most cited

H4 — Consistency & Standards
2nd most

H3 — User Control & Freedom
3rd most

Design Implication

→

We prioritized fixing system feedback, progress visibility, and consistent interactions.

- 📡 System feedback
- 📊 Progress visibility
- 🔁 Consistent interactions

Prioritizing Our Values and Goals

In-context meaning:
Reason Lab should reinforce reasoning before revealing assistance. Users must attempt, reflect, and revise before receiving support.

Solutions

1. Structured Explain-Back Requirement
Before unlocking a hint, users must submit a written explanation of their reasoning. The system evaluates effort before providing guidance.

- Encourages construction of understanding rather than passive answer consumption.

2. Layered Hint System
Hints are progressive (nudge → scaffold → deeper guidance) instead of revealing full solutions immediately.

- Maintains productive struggle while still supporting learning.

Summary of Evaluations

Active Learning

Problems with Hint Limit

- Hint limit can be bypassed by pressing Import after reaching the maximum.
- System does not consistently track the number of hints used.
- Makes the hint restriction ineffective and confusing for users.

Problems with Green Progress Bar

- Progress bar appears completely full even at 0/10 progress.
- Visual feedback contradicts the progress text.
- Users cannot accurately tell how close they are to unlocking hints.

-

Summary of Evaluations cont'd

Active Learning

Problems with System Feedback

- Actions like Submit, Import, and Clear instantly change text without feedback.
- No loading state, confirmation, or animation.
- Users cannot tell if the system processed their request.

Clear Button Issues

- Clear does not fully reset the interface.
- Hint counter and other system states remain unchanged.
- Leads to misleading progress and hint usage information.

-

-

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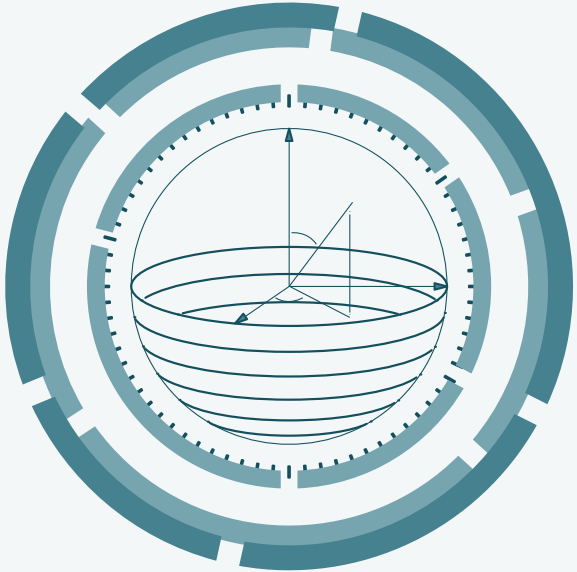
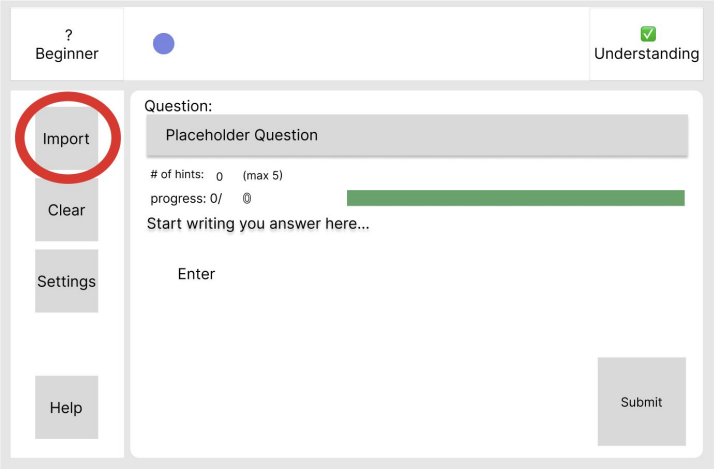
03

Major UI Revisions (Severity 3-4)



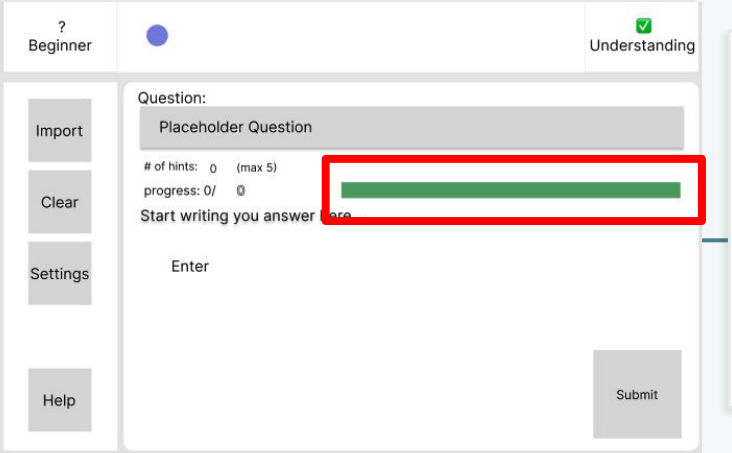
Inconsistent Consistency & Standards



No Confirmation After Import (H1)

After clicking "Import" and selecting "Input File," users reported that the interface immediately transitions to the question entry screen without confirming whether the file was successfully uploaded.

Violation: Import button does not follow standard file upload behavior. Clicking Import replaces text instead of opening a file browser.



Progress Bar Relationship Unclear (H1)

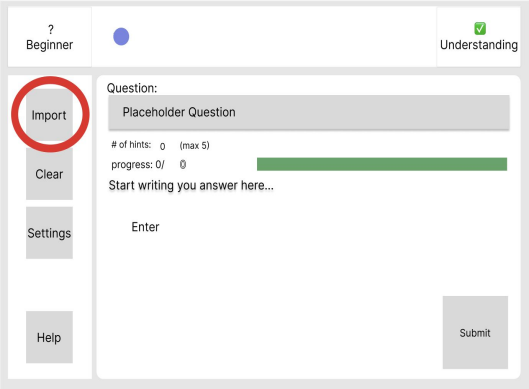
It is not clear to users how and why the sliding position on the top slider will affect the total progress bar length.

Violation: Currently there's no indication that as users set their progress to understandings, there are more steps they need to complete.

Major Change (Severity 4)

Fixing Import: Confirmation Flow

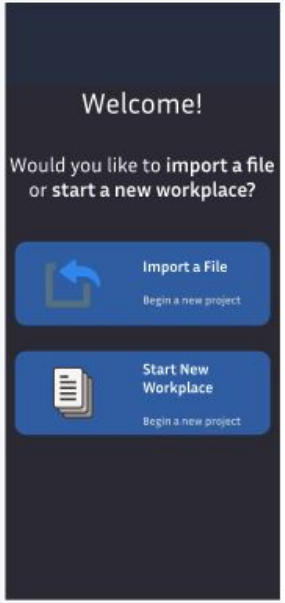
BEFORE



Fix: Clicking Import now opens a Welcome screen → native file browser → preview confirmation → analysis results. Users always know what step they're on.

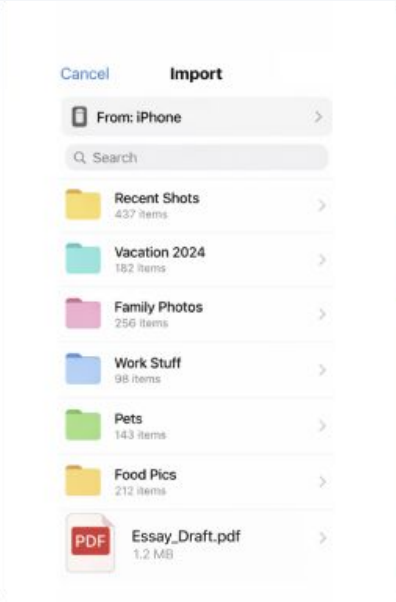
AFTER — 4-Step Confirmed Import Flow

1



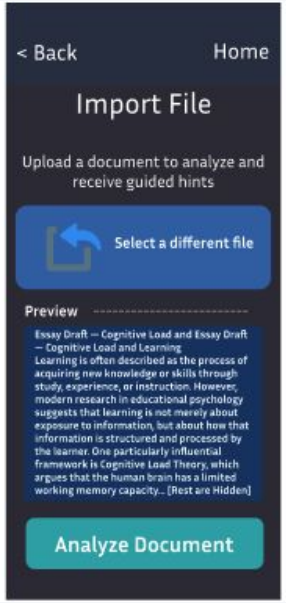
① Welcome

2



② File Picker

3



③ Import File

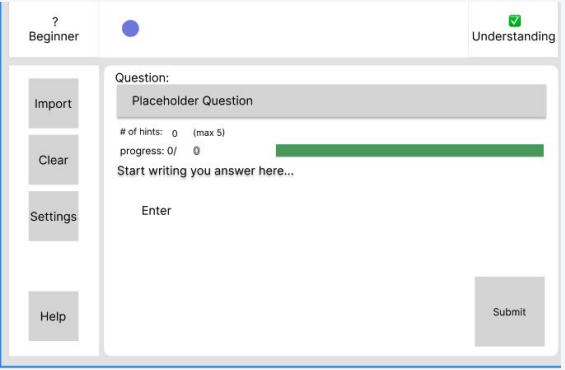
4

④ Results

Major Change (Severity 4)

Fixing Progress Bar

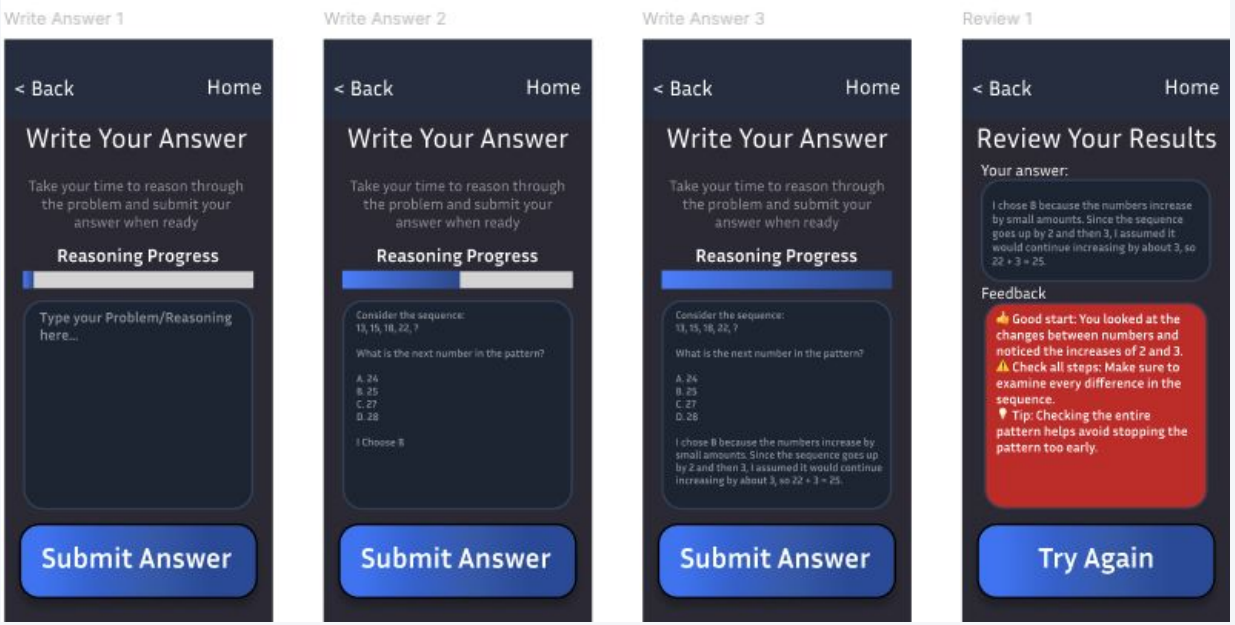
BEFORE



Fix: The Progress Bar now reflects how much explanation the user has written — giving real-time visual feedback on reasoning depth before submitting.

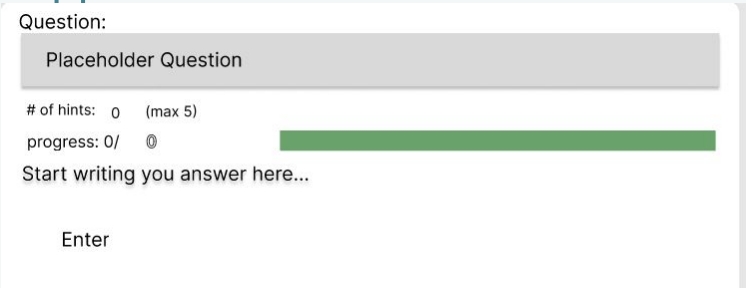
AFTER — 4-Step Confirmed Import Flow

1 2 3 4



① Welcome ② File Picker ③ Import File ④ Results

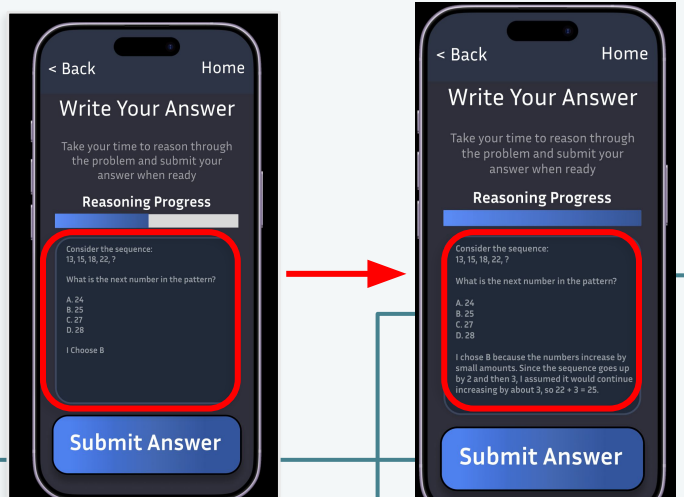
Problem w/ Progress Bar



Lack of Feedback + Error (H1/H5)

Users noticed that the progress bar, the primary feature displaying the user's progress would sometimes not grow/shrink with user progress. This made it difficult to tell when a user could submit an answer or whether they were making any progress at all.

Violation: Progress bar doesn't properly display progress, forming confusing visuals. Error + Visibility violations.



Update:

Ensured that a substantial increase in information lead to an increase in the reasoning progress. Additionally we scaled to each difficulty, how quickly the progress bar will fill.

Repairing the progress bar should make it much clearer to users exactly what their typed work is impacting. The speed the bar fills also helps display the difference in the difficulties better than before

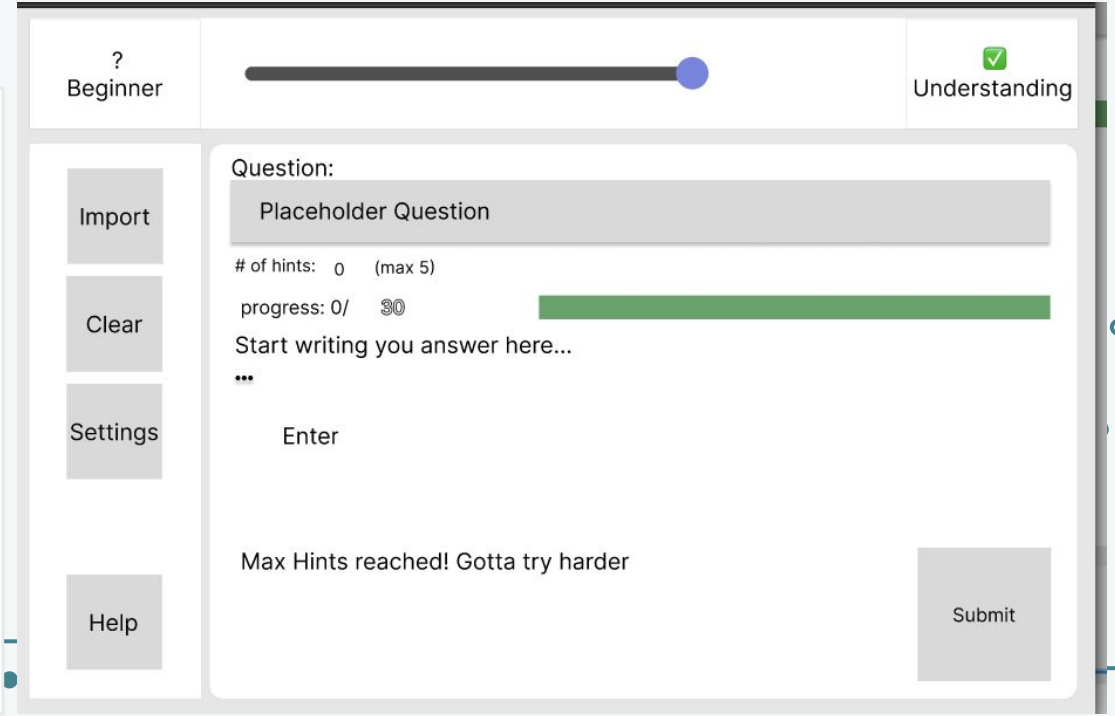
Visual Overhaul - BEFORE

Interactive Board

Our initial design was meant to be hands on, showing live updates as a user writes, with your difficulty sliders, hints, and progress all easily seen within a singular screen.

It also had a bunch of accessibility options through those extra sliders and settings. However, in practice, this resulted in a lot of confusion

- Input location was unclear
- Progress bar vs Slider blended
- # of Hints vs the Word Counter became cluttered



Visual Overhaul - AFTER

Mobile App

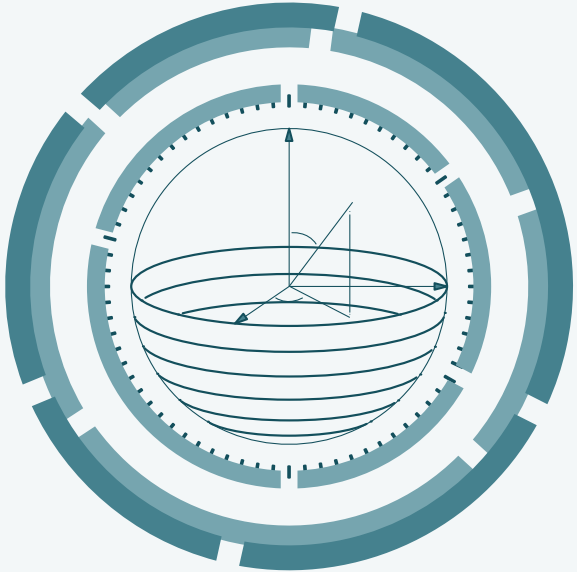
In our previous design, we noticed that a large number of our violations and confusions derived from the fact that all our info was compacted onto a single board. Even with additional side bars and openable tabs, it was still too much information for one base screen.

Due to all the previous complications, we began moving to an app

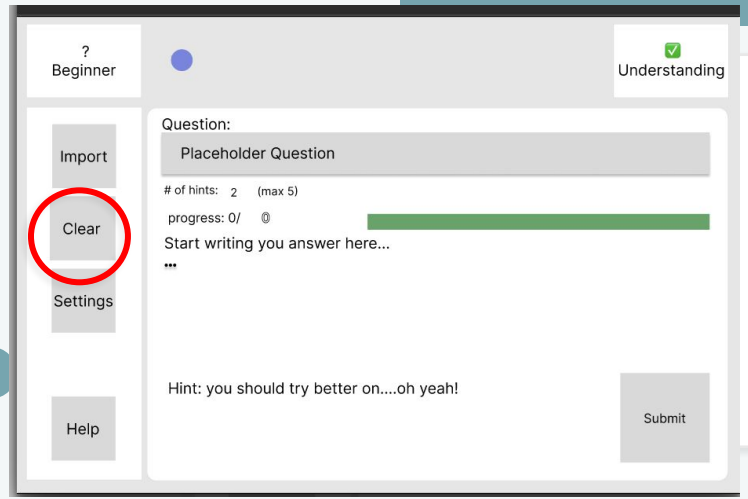
- Difficulties now selected through a screen
- Hints are now based on the difficulty level
- Importing a file now uses the mobile device's file system

04

Minor UI Revisions (Severity 1-2)



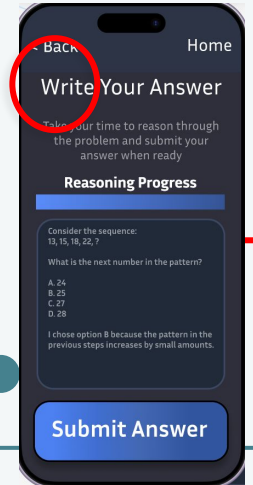
Clear Button Issues

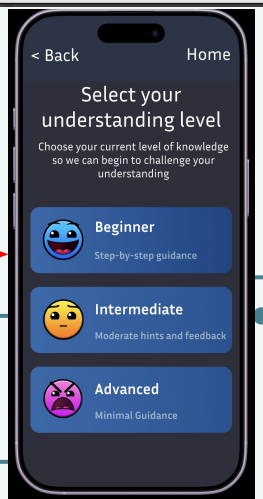


Clear Button Doesn't Actually Reset (H2), BEFORE UPDATE:

After clicking the clear button, the system fails to reset and 'clear' the remainder of the board. It is expected that the hint progress and all write categories are reset to being blank but the program fails to do so.

Violation: Clear button does not follow standard file upload behavior. Clicking Import replaces text instead of opening a file browser.

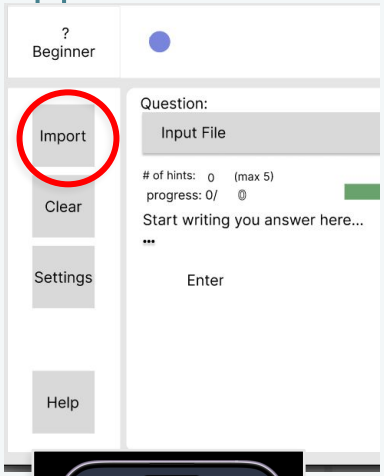


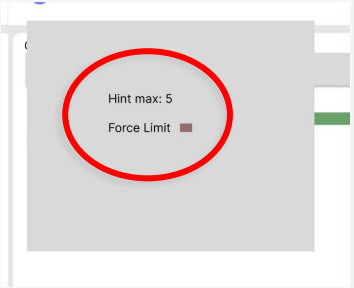


AFTER Update:

With the new format, a universal clear is no longer as necessary. Exchanged for back button to return a screen and a home button to head to the beginning of the app. This was to better match a standard app flow and does not run into functionality issues like that of the clear button. Additionally, using the back or home button allows the user to quickly reset their progress and opt for a new setting if the user so chooses.

Problem w/ System Feedback

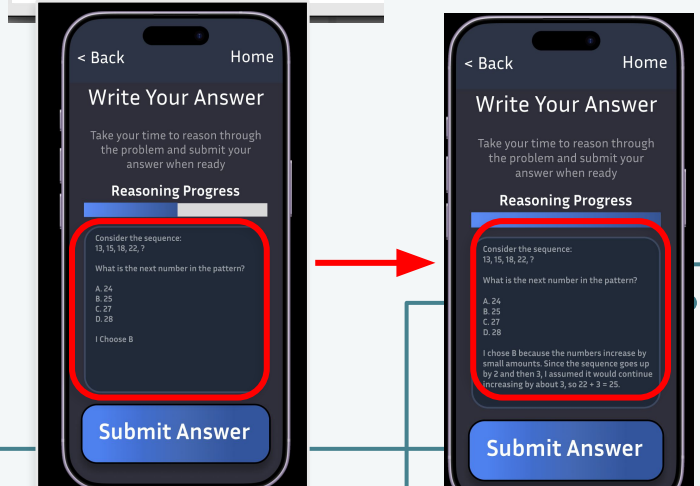




Lack of Confirmation/System Feedback (H1), BEFORE UPDATE:

Multiple users had complaints with the lack of visual feedback and user involvement throughout our prototype. The import file button lacked any sort of file selection and immediately imported an example question. This resulted in confusion regarding what even occurred in the prototype. Similar trends also occurred in the hint progression and the settings changes

Violation: Clear button does not follow standard file upload behavior. Clicking Import replaces text instead of opening a file browser.



AFTER Update:

Ensured that any area that would usually allow for user input displays an example user input, going step-by-step to display what a user could input. This should make it clearer at what the user is editing and what the app will allow them to change in the final prototype

Minor Fixes
(Severity 1–2)

Minor UI Revisions

Heuristic & Issue	Fix Applied	Impact on User Experience
01 H1 S2 **Visibility of System Status** The slider had no explanation of what it controlled — users couldn't tell if it affected difficulty, time, or scoring.	🏷️ **Fix Applied** Now, instead of a slider, a difficulty selection screen is provided with relevant descriptions.	→ **Impact** Users now immediately understand how the difficulty system works before interacting with it, reducing confusion and trial-and-error.
02 H1 S2 **Visibility of System Status** Hints appeared in the same plain white background as the rest of the workspace, making them visually indistinct from regular content.	💡 **Fix Applied** Added a dedicated hint screen with colored background highlight and border to the hint container, visually separating it from the main content area.	→ **Impact** Users can immediately recognize when a hint appears, improving awareness of new information and reducing the chance hints are missed.
03 H8 S1 **Aesthetic & Minimalist Design** Buttons did not change color or appearance when clicked or hovered, giving no visual confirmation that an interaction was registered.	🖱️ **Fix Applied** Added emphasis and interactions to all interactive buttons — On interactivity, interactive buttons flashes (indicating interaction) and the transformation is more visual.	→ **Impact** Improves the perceived responsiveness of the interface and reinforces that user actions are being recognized by the system.

Major Violations Addressed

FIXED

Heuristic	Issue	Severity	Fixed?
H1	No system feedback on import — interface transitions silently with no confirmation or loading state	S4	✅ Fixed
H4	Progress bar inconsistent — showed 0 steps and didn't update as users added reasoning	S4	✅ Fixed
H3	No navigation control — users had no back option or ability to undo actions mid-flow	S3	✅ Fixed
H5	Hint system bypass — users could skip required reasoning steps without warning or prevention	S3	✅ Fixed

DEFERRED

Violations Not Fixed			
H10	Help & documentation — no onboarding or contextual guidance	S2	⏸ Deferred
H7	Efficiency shortcuts — no filters, quick-access, or power-user features	S2	⏸ Deferred

Violations Fixed

4 / 6 S3–S4 issues

67% of critical violations resolved

Why Were Some Not Fixed?

H10 and H7 violations were identified but classified as lower priority. They fall outside the scope of this prototype iteration, which focused on core task flow and system feedback.

These will be addressed in the next design sprint.

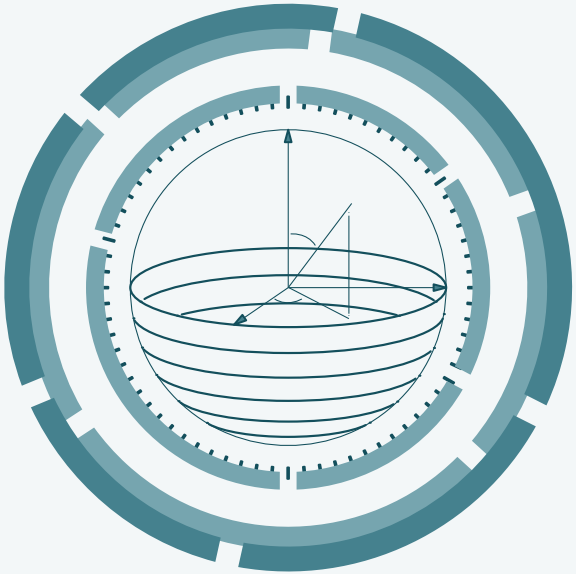
Progress Towards Usability Goals

How revised designs advance our core values, next steps & success metrics

	Impact on Design Values	Success Metric	Next Step
H4 — Consistency & Standards S4 Unified import flow: confirmed upload confirmation dialog replaces silent text swap	Users now trust the system behaves predictably. Reduces error-driven drop-off and aligns with standard file-upload conventions across iOS/Android.	**↓ Error rate on import task**	→ A/B test confirmation dialog copy to find lowest-friction wording
H4 — Consistency & Standards S4 Progress bar now reflects reasoning depth — updates dynamically as user writes	Users gain a sense of agency and completion. Supports the design value of transparency by making invisible effort visible in real time.	**↑ Task completion rate ↑ Avg. reasoning length**	→ Define threshold values for bar segments (e.g. 25 / 50 / 75 / 100 words)
H3 — User Control & Freedom S3 Phone-first layout adopted — taller viewport better suits the app's scroll-based interactions	Increases ecological validity of the prototype. Matches the mental model of users who primarily encounter study tools on mobile devices.	**↑ Perceived realism score in user testing**	→ Conduct usability test on physical device; verify tap target sizes meet 44px minimum

05

Impact on Usability Goals



Why We Switched to Phone Instead of Tablet

01

Inspired by Chatatouille

We took inspiration from the Chatatouille slides, which demonstrated that a phone-first layout creates a more focused and realistic user experience.

02

Mobile Phones Are More Common

Mobile phones are used far more frequently than tablets for everyday tasks. Designing for phones ensures we reach the widest audience.

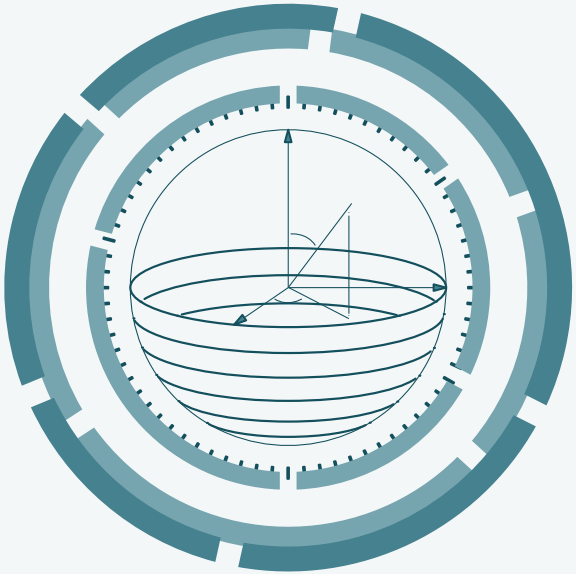
03

It Simply Looks Better

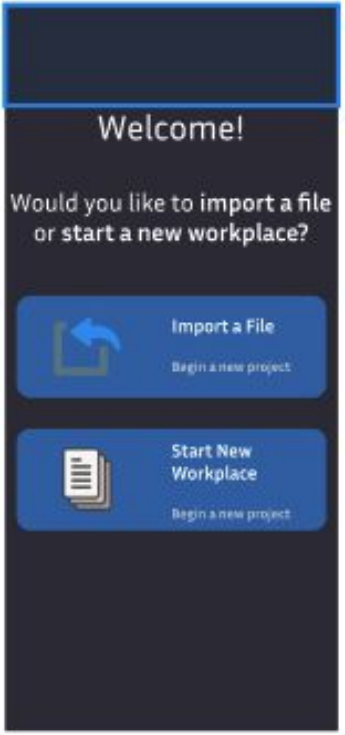
Phone screen dimensions produce a cleaner, more modern presentation. The taller aspect ratio better showcases the app's vertical scroll interactions.

06

Updated Prototype Designs

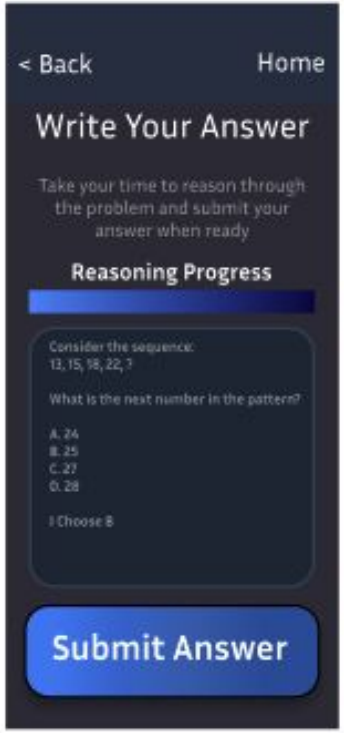


Simple


Welcome!
Would you like to import a file or start a new workplace?
Import a File
Begin a new project
Start New Workplace
Begin a new project

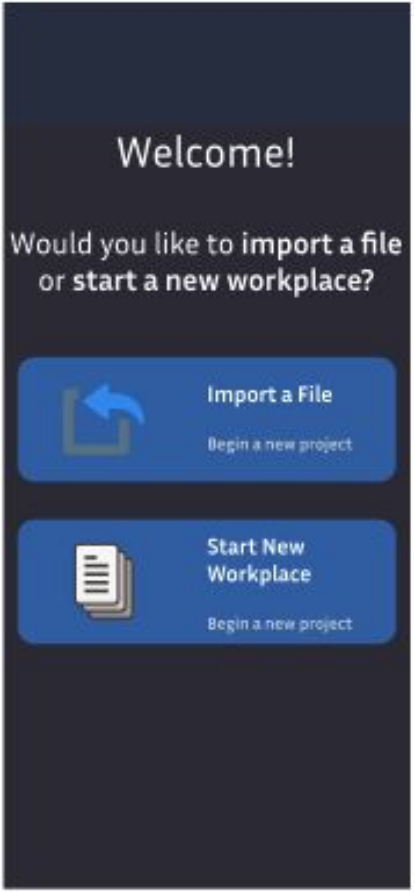
< Back
Home
Select your understanding level
Choose your current level of knowledge so we can begin to challenge your understanding
Beginner
Step-by-step guidance
Intermediate
Moderate hints and feedback
Advanced
Minimal Guidance

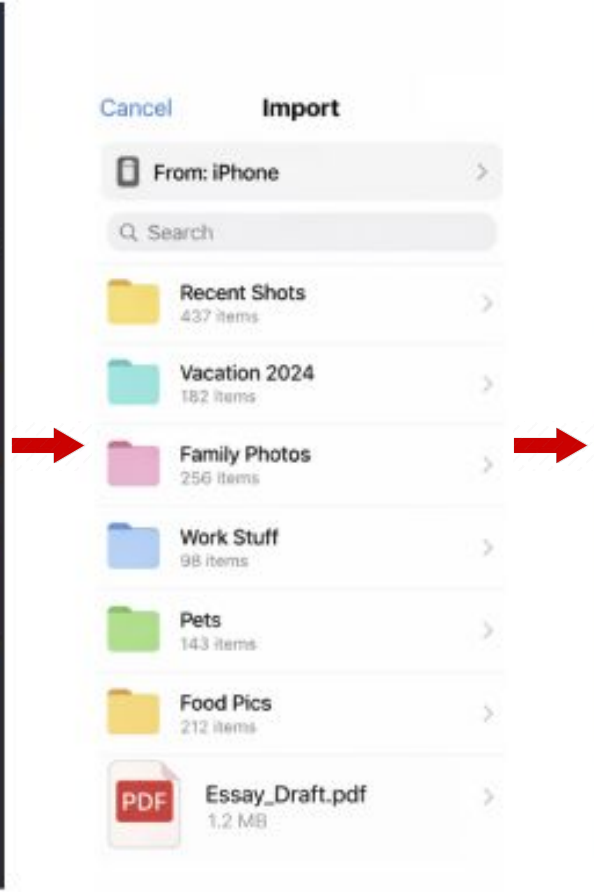
< Back
Home
Write Your Answer
Take your time to reason through the problem and submit your answer when ready
Reasoning Progress
Type your Problem/Reasoning here...
Submit Answer


< Back
Home
Write Your Answer
Take your time to reason through the problem and submit your answer when ready
Reasoning Progress
Consider the sequence:
13, 15, 18, 22, ?
What is the next number in the pattern?
A. 24
B. 25
C. 27
D. 28
I Choose B
Submit Answer

< Back
Home
Review Your Results
Your answer:
Consider the sequence:
13, 15, 18, 22, ?
What is the next number in the pattern?
A. 24
B. 25
C. 27
D. 28
I Choose B
Feedback
Your choice (B. 25) shows that you were trying to continue the pattern, which is good.
However, the pattern in this sequence follows a different rule than adding a constant number.
Step-by-step pattern
Look at the differences between each number:
- 15 − 13 = 2
- 18 − 15 = 3
- 22 − 18 = 4
The increases are 2, 3, 4, which go up by 1 each time.
So the next increase should be 5:
- 22 + 5 = 27
Correct answer: C. 27

Moderate


Welcome!
Would you like to import a file or start a new workplace?
Import a File
Begin a new project
Start New Workplace
Begin a new project


Cancel
Import
From: iPhone
Search
Recent Shots
437 items
Vacation 2024
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Family Photos
256 items
Work Stuff
98 items
Pets
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Food Pics
212 items
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Essay_Draft.pdf
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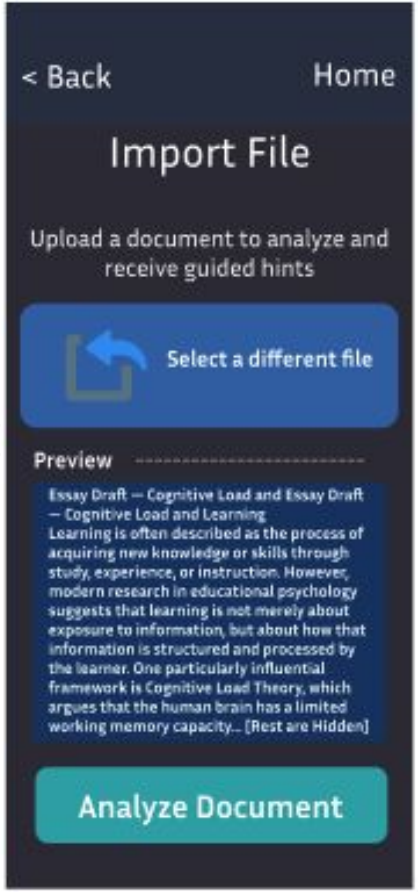
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Import File
Upload a document to analyze and receive guided hints
Select a different file
Preview
Essay Draft — Cognitive Load and Essay Draft — Cognitive Load and Learning
Learning is often described as the process of acquiring new knowledge or skills through study, experience, or instruction. However, modern research in educational psychology suggests that learning is not merely about exposure to information, but about how that information is structured and processed by the learner. One particularly influential framework is Cognitive Load Theory, which argues that the human brain has a limited working memory capacity... [Rest are Hidden]
Analyze Document

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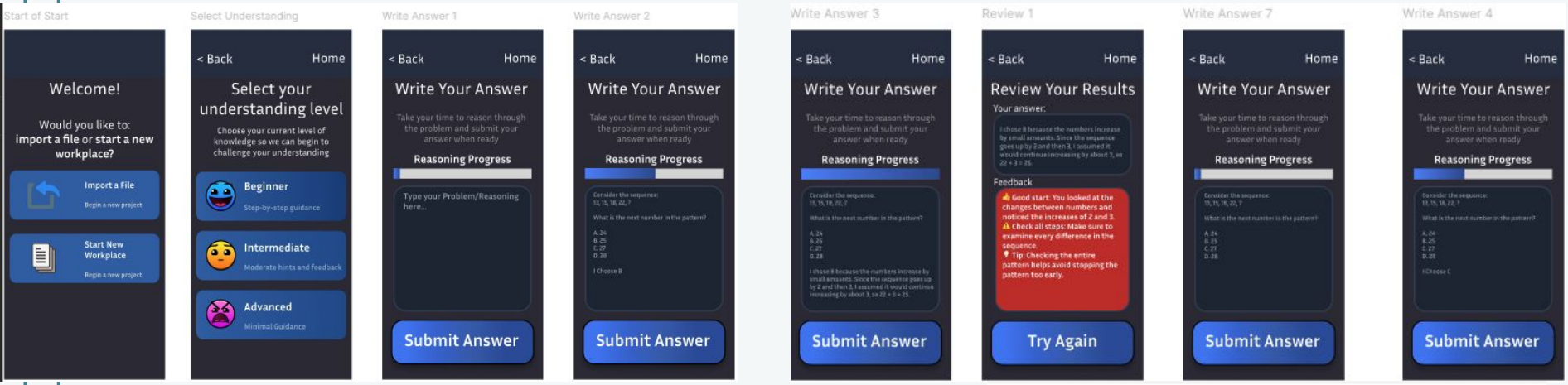
Recommendations
Preview
Essay Draft — Cognitive Load and Essay Draft — Cognitive Load and Learning
Learning is often described as the process of acquiring new knowledge or skills through study, experience, or instruction. However, modern research in educational psychology suggests that learning is not merely about exposure to information... [Rest are Hidden]
Hints
Your introduction explains the topic but does not clearly state your main argument or thesis. Consider adding one sentence that summarizes what you will argue about cognitive load and learning.
The phrase "modern research in educational psychology suggests" is vague. Try referencing a specific researcher, study, or example to strengthen this claim.
You mention working memory limitations, but you could briefly explain why the brain struggles with too much information at once.

Complex


Start of Start
Welcome!
Would you like to:
import a file or start a new workplace?
Import a File
Begin a new project
Start New Workplace
Begin a new project
Select Understanding
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Home
Select your understanding level
Choose your current level of knowledge so we can begin to challenge your understanding
Beginner
Step-by-step guidance
Intermediate
Moderate hints and feedback
Advanced
Minimal Guidance
Write Answer 1
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Write Your Answer
Take your time to reason through the problem and submit your answer when ready
Reasoning Progress
Type your Problem/Reasoning here...
Submit Answer
Write Answer 2
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Home
Write Your Answer
Take your time to reason through the problem and submit your answer when ready
Reasoning Progress
Consider the sequence:
13, 15, 18, 22, ?
What is the next number in the pattern?
A. 24
B. 25
C. 27
D. 28
I Choose B
Submit Answer
Write Answer 3
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Home
Write Your Answer
Take your time to reason through the problem and submit your answer when ready
Reasoning Progress
Consider the sequence:
13, 15, 18, 22, ?
What is the next number in the pattern?
A. 24
B. 25
C. 27
D. 28
I chose B because the numbers increase by small amounts. Since the sequence goes up by 2 and then 3, I assumed it would continue increasing by about 3, so 22 + 3 = 25.
Submit Answer
Review 1
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Home
Review Your Results
Your answer:
I chose B because the numbers increase by small amounts. Since the sequence goes up by 2 and then 3, I assumed it would continue increasing by about 3, so 22 + 3 = 25.
Feedback
Good start: You looked at the changes between numbers and noticed the increases of 2 and 3.
Check all steps: Make sure to examine every difference in the sequence.
Tip: Checking the entire pattern helps avoid stopping the pattern too early.
Try Again
Write Answer 7
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Home
Write Your Answer
Take your time to reason through the problem and submit your answer when ready
Reasoning Progress
Consider the sequence:
13, 15, 18, 22, ?
What is the next number in the pattern?
A. 24
B. 25
C. 27
D. 28
Submit Answer
Write Answer 4
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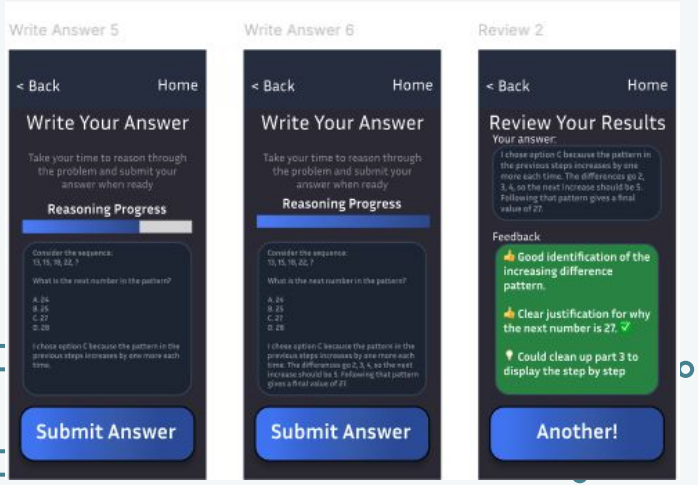
Home
Write Your Answer
Take your time to reason through the problem and submit your answer when ready
Reasoning Progress
Consider the sequence:
13, 15, 18, 22, ?
What is the next number in the pattern?
A. 24
B. 25
C. 27
D. 28
I Choose C
Submit Answer


Write Answer 5
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Home
Write Your Answer
Take your time to reason through the problem and submit your answer when ready
Reasoning Progress
Consider the sequence:
13, 15, 18, 22, ?
What is the next number in the pattern?
A. 24
B. 25
C. 27
D. 28
I chose option C because the pattern in the previous steps increases by one more each time.
Submit Answer
Write Answer 6
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Home
Write Your Answer
Take your time to reason through the problem and submit your answer when ready
Reasoning Progress
Consider the sequence:
13, 15, 18, 22, ?
What is the next number in the pattern?
A. 24
B. 25
C. 27
D. 28
I chose option C because the pattern in the previous steps increases by one more each time. The differences go 2, 3, 4, so the next increase should be 5. Following that pattern gives a final value of 27.
Submit Answer
Review 2
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Home
Review Your Results
Your answer:
I chose option C because the pattern in the previous steps increases by one more each time. The differences go 2, 3, 4, so the next increase should be 5. Following that pattern gives a final value of 27.
Feedback
Good identification of the increasing difference pattern.
Clear justification for why the next number is 27.
Could clean up part 3 to display the step by step
Another!

06

Figma